# 📘 SYSTEM DESIGN DOCUMENT (UML)

**ZJNU_HACKATHON_2025_TEAM10**

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## 1. Introduction

This System Design Document explains the internal architecture of **Handy** & **HandyChat** using UML diagrams.
It covers:

- Use Case Diagram
- System Architecture Diagram
- Class Diagram
- ER Diagram (Database)
- Sequence Diagram
- Data Flow Diagram

These diagrams illustrate how the frontend, backend, services, AI engine, and MySQL interact.

# 2. Use Case Diagram

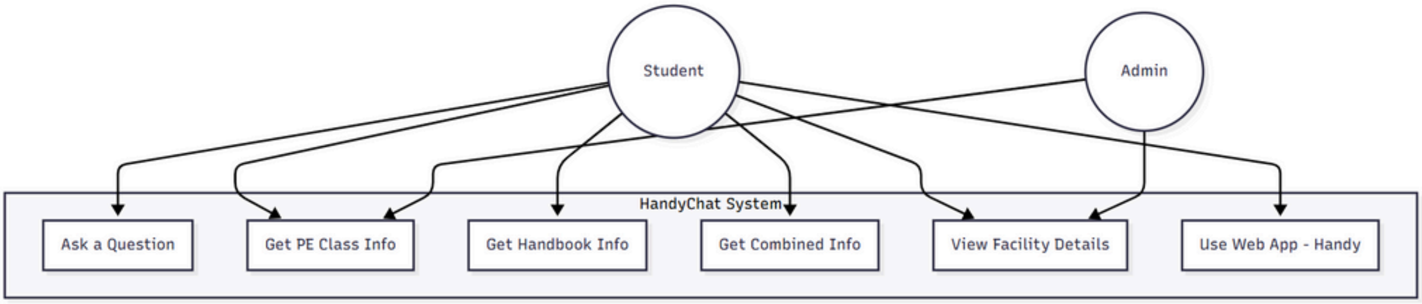
### Actors

- **Student** (primary user)
- **Admin / Staff** (future use, currently minimal)
- **AI Service** (system component)
- **Database** (system component)

### Use Cases

- Ask a question
- View PE class locations
- View schedules
- Read sports facility info
- Ask handbook questions
- Ask combined questions
- Access web interface (Handy)

### ✅ Use Case Diagram (Mermaid UML)

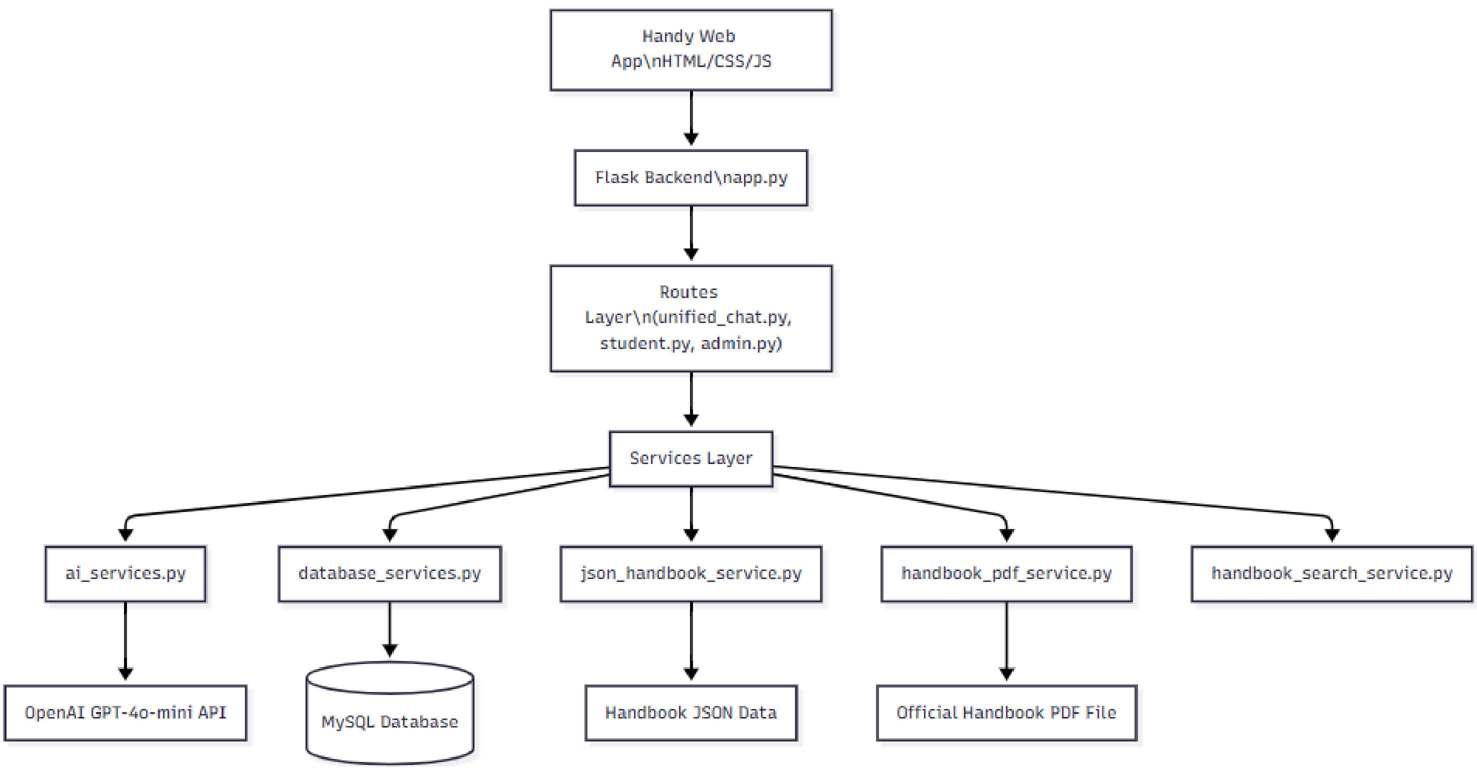



# 3. High-Level Architecture Diagram

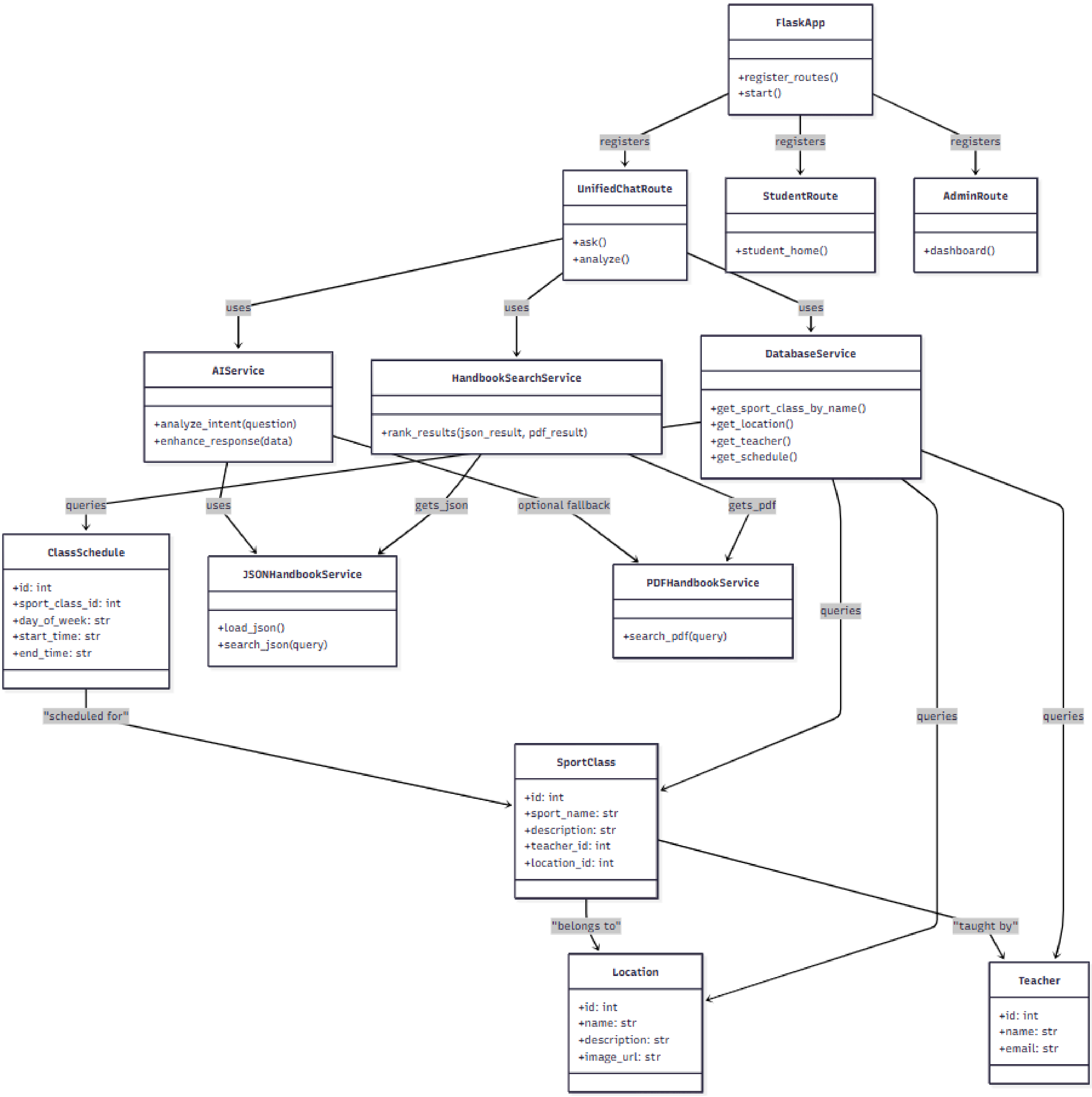
Components:

- **Frontend (Handy)** — HTML, CSS, JS
- **Flask Backend** — app.py
- **Routes Layer** — unified_chat, admin, student
- **Services Layer** — AI + Database + Handbook Search
- **MySQL** — PE tables
- **JSON/PDF Handbook** — non-DB content
- **OpenAI** — AI inference

### ✅ System Architecture Diagram




# 4. Class Diagram (Backend Core Classes)



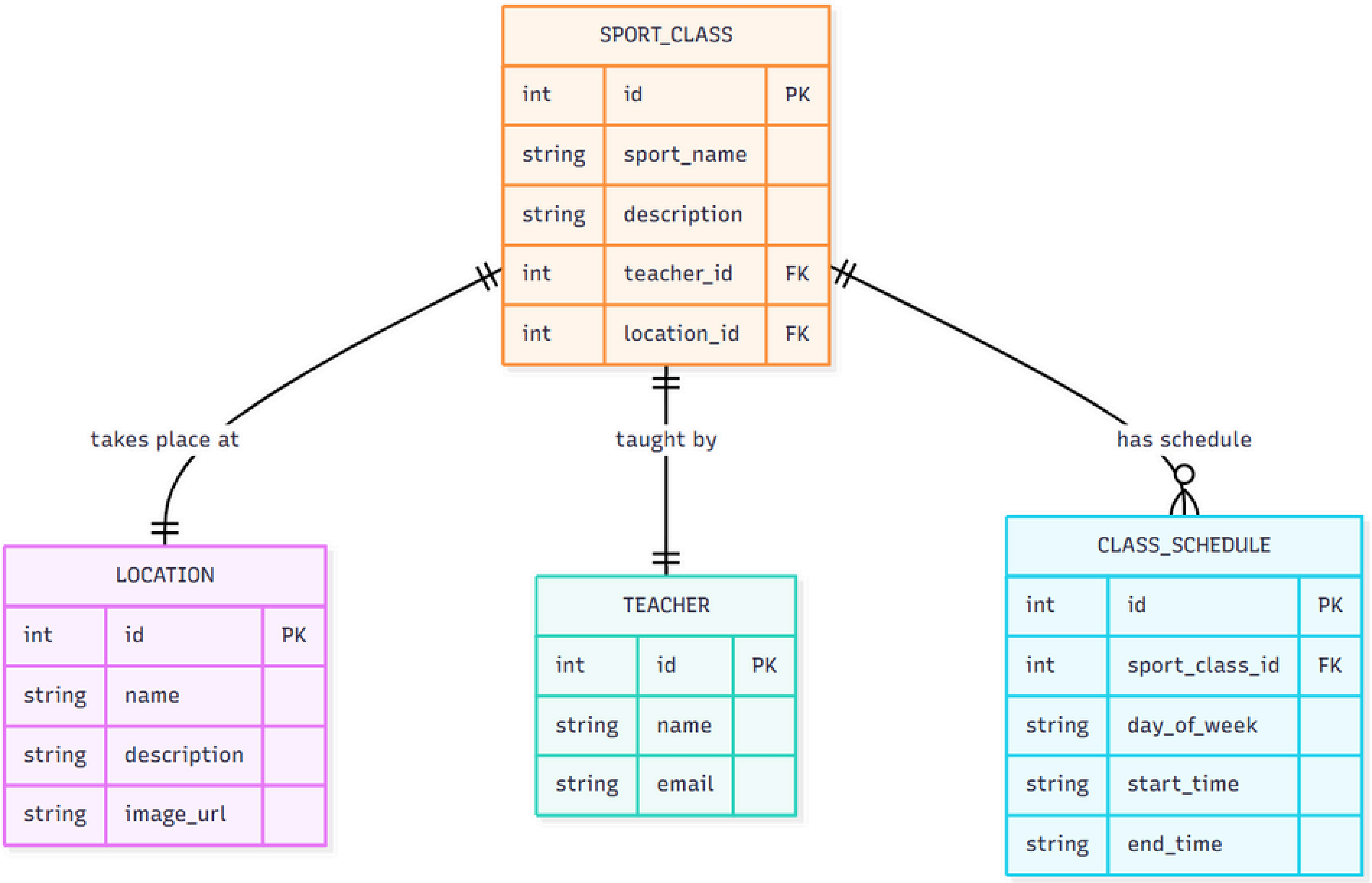

# 5. ER Diagram (Database Schema)

(Updated — handbook tables removed!)

Tables:

- **location**
- **teacher**
- **sport_classes**
- **class_schedule**

## ✅ ER Diagram

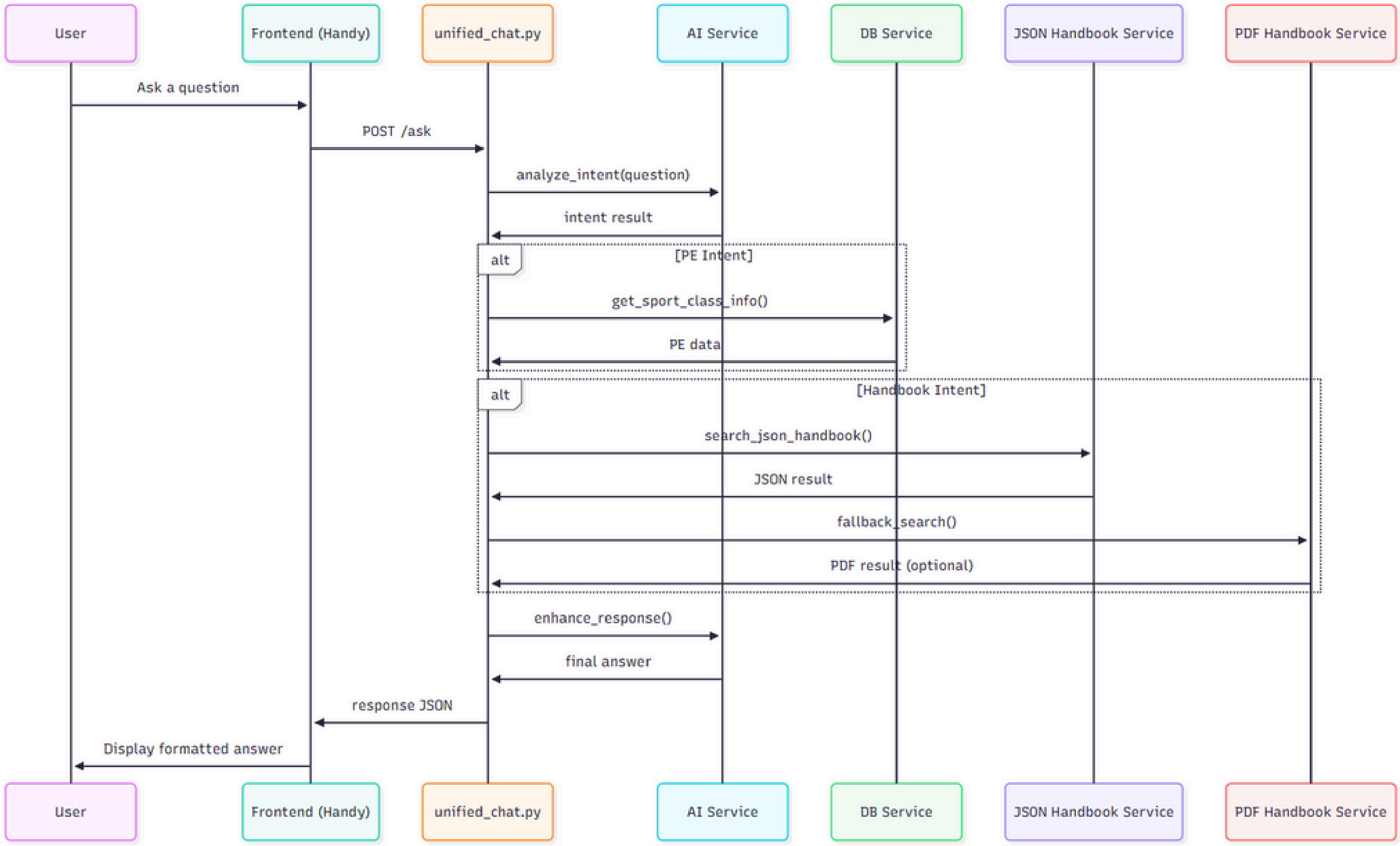



# 6. Sequence Diagram (Main Chat Request Flow)

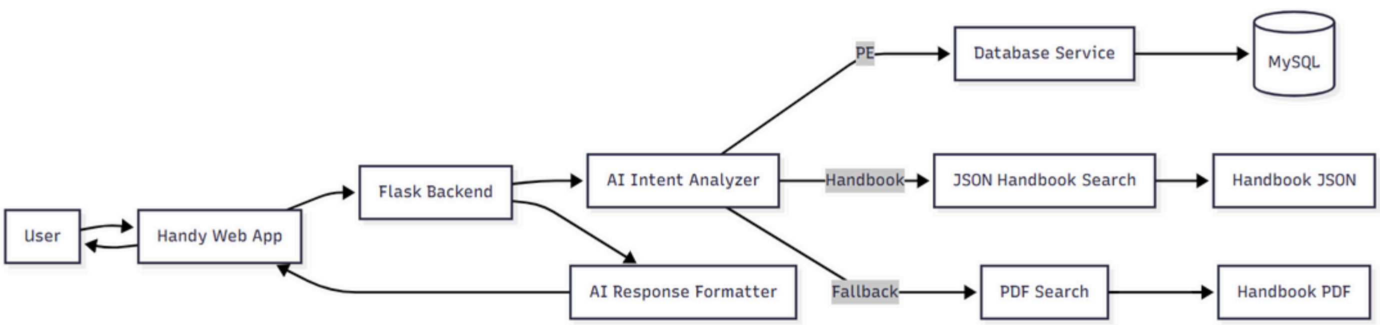
Steps:

1. Student sends a question.
2. unified_chat receives it.
3. AI analyzes intent.
4. The system routes to:
    - PE lookup
    - Handbook JSON search
    - Handbook PDF fallback
5. Combine results
6. AI enhances the answer
7. Return response to frontend

### ✅ Sequence Diagram




# 7. Data Flow Diagram (DFD – Level 1)




# 8. System Components Description

## 8.1 Frontend

- Uses HTML templates
- Styled with CSS
- JS handles chat UI interaction

## 8.2 Backend (Flask)

- Accepts all chat requests
- Routes to correct services
- Coordinates AI + DB + JSON/PDF

### 8.3 Services Layer

Separate Python modules for:

- AI
- MySQL operations
- Handbook JSON search
- Handbook PDF search
- Combined search ranking

### 8.4 Database

Stores **only PE data**.

### 8.5 Handbook Layer

- JSON data = fast, structured searches
- PDF fallback = deep, flexible search

# 9. Future System Design Improvements

- Admin dashboard for updating PE classes
- Database migration for storing handbook content
- Caching layer for faster handbook search
- WebSocket communication for real-time typing indicators
- WeChat Mini Program version
- Mobile app version

# 10. Conclusion

This System Design Document provides a complete visual and structural understanding of the Handy & HandyChat system.
It includes all major UML diagrams judges and developers expect in a professional project submission.